Final audited research report — F31
Quality-Controlled EMG Filtering and Bilateral Temporal-Progression Classification

Report information

Field	Value
Study identifier	F31 — local EMG analysis
Report title	Quality-Controlled EMG Filtering and Bilateral Temporal-Progression Classification
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Public code repository	https://github.com/EthanGL1/proyecto-biomecanica-emg
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Abstract

Surface electromyography (sEMG) can describe muscle electrical activity but does not independently establish physiological fatigue. This study audited 140 local recordings from 15 recording-derived groups and compared temporal-position classifiers using only the recordings. The participant spreadsheet and external datasets were excluded. R1–R5 contributed 138 recordings and 414 segments labelled as the early, middle or late elapsed-time third; R0/R0_1 were quality-control-only. Five filter branches, seven feature-family combinations and 11 classifier families (325 fixed configurations) were evaluated using 15 outer leave-one-participant-out folds and four grouped inner folds. The global nested selection procedure achieved participant-mean macro-F1 0.740 (SD 0.099; 95% participant-bootstrap CI 0.689–0.787) and macro ROC-AUC OvR 0.941 (SD 0.038; CI 0.921–0.959). Logistic regression had the highest fixed-family macro-F1 point estimate, 0.751 (SD 0.108), without evidence of superiority. Post hoc stratification found left/right macro-F1 values of 0.776 and 0.686, respectively; the paired 14-participant interval included zero. MDF and MNF showed predominantly negative time-associated slopes, while RMS had a positive median slope, but recordings were non-monotonic and lacked synchronized force, verified repetition events and independent fatigue labels. All confidence intervals resampled participants, preserving dependence among each person's recordings and segments. The work therefore supports a reproducible temporal-position benchmark and filter audit, not a fatigue detector, physiological threshold, onset or peak.

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Keywords

surface electromyography; signal quality; digital filtering; temporal-position proxy; grouped validation; bilateral analysis

1. Fatigue: definition and state of the art

1.1 Which fatigue are you detecting?

The outcome in this study is not central or peripheral fatigue, nor a verified acute or accumulated fatigue state. It is a three-class proxy for elapsed recording position: early, middle and late thirds within each eligible R1–R5 file. The owner reported that exercise continued to technical failure, but no per-file failure timestamp, verified repetition boundary, force trace or load record was available to anchor the labels. A model that predicts these thirds has not thereby detected fatigue.

1.2 Physiological basis

During a sustained contraction at approximately controlled force, changes in recruitment and motor-unit firing, muscle-fibre conduction velocity and spectral power distribution can accompany fatigue. A downward shift in median or mean frequency is a commonly described pattern under those constraints; RMS can increase, remain stable or decrease as recruitment, effort and force capacity change. Dynamic movement, changing force, electrode motion, sweat, temperature and signal conditioning can alter the same descriptors [1], [31], [32], [34], [38], [39]. For a centered segment x_c[n], RMS = sqrt((1/N) sum x_c[n]^2). For power spectrum P(f), MNF = sum_f fP(f)/sum_f P(f), and MDF is the frequency at which cumulative power reaches one half of total power over the analysis band. A negative spectral slope can be compatible with fatigue in controlled protocols but is not a universal threshold or proof of fatigue.

1.3 How fatigue has been detected from EMG - review

TABLE I. COMPARATIVE PRIMARY STUDIES ON sEMG FATIGUE OR RELATED PROGRESSION TASKS

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Reference	Task / sample	Target and labelling	Features / model	Validation	Reported result and comparability limit
[3] Liu et al. (2021)	Dynamic sEMG fatigue classification; task-specific protocol	Study-defined fatigue states; not the file-time target used here	EMG descriptors with optimized SVM	SVM	Published score not re-pooled; labels and split are not directly comparable to participant-wise thirds.
[5] Wang et al. (2021)	Cycling; 20 men	Anaerobic-threshold protocol	Wavelet-packet denoising; RMS/IEMG/frequency features; LSTM	LSTM	Reported held-out accuracy 0.9518; participant grouping was not established in reviewed notes.
[6] Mu et al. (2022)	Exercise task; 30 adults	Three RPE-based states	sEMG+ECG features; temporal convolutional network	TCN	Random 10-fold reported; subject grouping not established in reviewed notes.
[11] Otálora et al. (2023)	Biceps curl; 30 participants	Borg CR10 and sensor protocol	sEMG, IMU and optical force sensing; LightGBM	LightGBM	Reported 96.2% multimodal and 78.8% EMG-only; test-participant assignment incomplete in reviewed notes.
[14] Mu et al. (2024)	Cycle ergometry; 80 participants	Protocol/time and anaerobic-threshold criteria	sEMG+accelerometry; attention CNN	CNN with attention	Reported accuracy 92.3–92.52% and F1 91.92–92.29%; grouping unit unclear in reviewed notes.
[15] Zhang et al. (2024)	Lower-limb FES-sEMG; 5 participants	Borg-based states	500 ms STFT/CWT/HHT image features; CNN/LSTM	CNN/LSTM	HHT class F1 0.911, 0.884 and 0.876; random-window 8:1:1 does not establish participant independence.
[23] Hwang et al. (2025)	Gait after step-ups; 35 adults	RPE ≥17 protocol endpoint	sEMG and IMU; CNN-LSTM-attention	Fusion and sEMG-only	LOSO reported; estimates RPE/protocol fatigue, not local physiological events.
[25] Huh et al. (2026)	Preacher curls over five days; 17 adults	Normalized onset/progression proxy	sEMG e-tattoo and BH-IEEMD processing	Progression estimator	LOSO with per-person/day calibration; not an independent fatigue ground truth.
[26] Zhai et al. (2026)	30 s sprint; elite cyclists	Relative power-drop target	sEMG deep CNN-BiLSTM-attention	Continuous estimator	Reported MAE 0.048±0.019, R^2 0.493±0.249 and r 0.822±0.123; calibration used held-out-person windows.
[34] da Silva et al. (2008)	Dynamic trunk flexion/extension to exhaustion	Protocol-defined exhaustion	Fourier vs. wavelet EMG indices	Validity/reliability comparison	Reported correlations 0.06–0.68 and ICC 0.00–0.88; dynamic indices showed variable validity/reliability.

The literature compares protocol-defined fatigue, exertion, mechanical power loss and temporal progression using different sensors and validation designs. Studies include EMG feature classifiers and dynamic tasks [2]–[4], cycling or multimodal exertion labels [5]–[7], simulated fatigue and human–robot applications [8]–[10], wearable systems and sensor fusion [11]–[16], dynamic gripping, recovery and multimodal classifiers [17]–[24], and multi-day or

continuous progression estimates [25]–[27]. The comparative table records target, population, features, model, validation and comparability limits. Published scores cannot be ranked directly against this study because its labels are file-time thirds, not independent force, Borg, power, threshold or fatigue-event labels [18], [27], [33].

1.4 Gap and contribution

This work does not claim a new fatigue detector. Its contribution is a traceable quality audit, mathematical comparison of five digital filter branches, grouped nested evaluation of a time-position proxy, post hoc left/right and R1–R5 summaries, and a cautious analysis of RMS/MDF/MNF trends. Stratification uses already frozen out-of-fold predictions and does not reselect models by side or series. Neither the side comparison nor the signal trends establish physiological asymmetry.

2. Dataset and acquisition

2.1 Protocol as executed

The audit covered 140 local files with 4,256,257 samples and 15 participant groups inferred from recording identifiers. The naming convention was confirmed by the user: iz denotes left, dr denotes right, and R1–R5 are the five series; R0/R0_1 are preliminary/noisy or anomalous takes reserved for QC. The eligible benchmark contained 138 R1–R5 files (70 left, 68 right) and 414 equal-time-third segments. The participant spreadsheet was not opened or used. Hardware, electrode site, muscle identity, analog chain, MVC, load, cadence, rests, physical calibration and per-recording repetition count remain unavailable. The column called Voltage_mV is an algebraic transformation of ADC counts and is not a verified electrode-voltage calibration.

2.2 Participants

TABLE II. AUDITED LOCAL DATASET AND ANALYSIS COVERAGE

Variable	Count / summary	Range / note
Recording-derived participant groups	15	Codes only; no demographic CSV used
Local EMG files audited	140	4,256,257 samples in total
R1–R5 benchmark recordings	138	70 left; 68 right
Temporal-proxy segments	414	138 early, 138 middle, 138 late

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Variable	Count / summary	Range / note
R0 / R0_1 quality-control files	2	QC only; not model training
Demographics / body mass / height	Not analyzed	participants.csv explicitly excluded
Task / muscle site / load	Not verified	File-level linkage and acquisition metadata absent

2.3 Exclusions

Structural exclusion criteria were non-finite signal or time values, unequal vectors, non-increasing timestamps that could not be split, or unrecoverable corruption. The QC audit found no structural exclusion. Timestamp discontinuities were defined by $\Delta t > \max(1.5\ \text{ms}, 1.5 \times \text{median positive } \Delta t) + 1\ \text{ns}$; they were treated as boundaries and not interpolated. No recording was excluded because of model performance. The two R0/R0_1 files were kept for QC and excluded from the benchmark by their recording class.

2.4 Signal quality assessment

The QC audit found 246 timestamp gaps across 41 files (241 in R1–R5 files and five in the two QC-only files). It recorded 74 large-derivative review flags, which are not diagnoses of movement artifact. No file crossed the predeclared 10 dB 50/60 Hz line-prominence screen, so no notch filter was applied. SNR was not estimated because a clean or resting reference was unavailable. ADC rails were undocumented; saturation therefore could not be confirmed or ruled out. Signal mode (RAW, rectified or envelope), electrode placement and analog acquisition filtering remain unknown.

3. Labelling strategy

3.1 Chosen strategy and justification

The frozen target divides each eligible recording into three equal elapsed-time intervals. Every R1–R5 file yields one early, one middle and one late segment, producing 138 segments per class and 414 total. This deterministic rule is reproducible but has no independent physiological meaning. R0/R0_1 have no predictive labels. Repetition-based or failure-event labels were not fabricated because synchronized event markers were absent.

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3.2 Class definition and distribution

TABLE III. FROZEN TEMPORAL-POSITION LABELS

Class	Exact rule	Segments (n)	%
Early	First third of elapsed recording time	138	33.33%
Middle	Second third of elapsed recording time	138	33.33%
Late	Final third of elapsed recording time	138	33.33%
Total	Three deterministic thirds per R1–R5 recording	414	100.00%

3.3 Circularity check

Label construction used elapsed time alone; the feature families contain waveform descriptors rather than the target label or a direct time-third field. Predictor names and target-generation logic are separate in the local pipeline. This prevents direct label-column leakage but does not remove association between descriptors and time, contraction sequence, pauses or recording duration. The result must still be interpreted as temporal-position classification.

3.4 Label reliability

The labels are generated deterministically, so inter-rater agreement is not applicable. No independent repetition, force, torque, Borg rating or failure event is linked to segments; label reliability against a fatigue construct is untested. There is no independent local fatigue ground truth and no external validation in the current final study, consistent with the project's local-only instruction.

4. Signal processing and feature extraction

4.1 Preprocessing chain

TABLE IV. DIGITAL FILTER CANDIDATES AND SIGNAL-HANDLING RULES

Stage	Method	Parameters and reason
Stored channel control	No digital filter	Comparator; storage mode not verified
High-pass IIR	4th-order Butterworth, 20 Hz	Forward–backward per contiguous run
High-pass IIR	4th-order Butterworth, 40 Hz	Forward–backward per contiguous run

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Stage	Method	Parameters and reason
Band-pass IIR	4th-order Butterworth, 20–400 Hz	Forward–backward per contiguous run
Band-pass FIR	101 taps; Hamming; 20–400 Hz	Reflective edge handling
Power-line notch	Not applied	No file reached the predeclared 10 dB 50/60 Hz prominence screen
Gaps / edges	Split at max(1.5 ms, 1.5×median Δt); common edge mask	No interpolation; incomplete spectral windows omitted
Rectification / normalization	Not applied	Stored mode and MVC reference unavailable; no physical-voltage calibration

Five branches were compared on the stored channel: control; fourth-order Butterworth high-pass at 20 Hz (HP20); fourth-order Butterworth high-pass at 40 Hz (HP40); fourth-order Butterworth band-pass 20–400 Hz; and a 101-tap (order 100) Hamming-window FIR band-pass 20–400 Hz. Timestamp-derived sampling frequency was approximately 1 kHz; actual coefficients were recomputed for each contiguous run. For an order-N Butterworth prototype, $|H_a(j\Omega)|^2 = 1/[1+(\Omega/\Omega_c)^{(2N)}]$ for the low-pass prototype, with high-pass/band-pass transformations and bilinear prewarping $\Omega=2f_s \tan(\pi f/f_s)$. The implemented IIR cascade is $H(z)=\Pi_s (b0,s+b1,s\, z^{-1}+b2,s\, z^{-2})/(a0,s+a1,s\, z^{-1}+a2,s\, z^{-2})$, with normalized $a0,s=1$; the difference equation is $y[n]=\Sigma_k b_k x[n-k]-\Sigma_{k\geq1}a_k y[n-k]$. The FIR response is $H(z)=\Sigma_{n=0}^{100}b_n z^{-n}$, where b_n is the ideal 20–400 Hz sinc difference multiplied by $w[n]=0.54-0.46\cos(2\pi n/100)$ and normalized at the band centre. The 1 kHz reference coefficients are supplied in the coefficient CSV; they are not substitutes for the run-specific coefficients.

4.2 Features computed

TABLE V. INTERPRETABLE FEATURE FAMILIES

Feature	Family	Definition	Interpretive reason and limit
Centered RMS	Amplitude / time	sqrt(mean(xc²))	Tracks signal amplitude; may rise, plateau or fall; nonspecific
Median frequency (MDF)	Spectrum	Half cumulative PSD power, 20–400 Hz	May shift with conduction and recruitment; also task/filter dependent
Mean frequency (MNF)	Spectrum	sum(f·P)/sum(P), 20–400 Hz	Summarizes spectral power distribution; not a standalone fatigue marker
Peak frequency	Spectrum	Frequency of maximum PSD bin	Sensitive to spectral shape and noise; exploratory
Normalized spectral entropy	Spectrum	Normalized entropy of 20–400 Hz PSD	Describes spectral dispersion; no unique physiological direction
Relative band powers	Spectrum	Power in 20–60, 60–150, 150–250, 250–400 Hz / total	Describes frequency redistribution; sensitive to band/filter choice
Waveform descriptors	Time-domain	Distribution, amplitude quantiles, line length and crossings	Characterize observed waveform; affected by contraction and acquisition
Between-window IQR	Inter-window variability	IQR of RMS and spectral descriptors in contiguous windows	Quantifies local variability; not a wavelet or validated fatigue index

IIR filters were applied forward and backward to each contiguous run with reflect padding; FIR used symmetric convolution with reflect padding and a 50-sample edge mask. No run was bridged across a gap and no interpolation, rectification, MVC normalization or conversion to physical voltage was applied. For the interior of a forward–backward IIR, $H_{fb}(e^{j\omega})=H(e^{j\omega})H(e^{-j\omega})$, giving zero net phase and magnitude $|H|^2$; a one-pass Butterworth −3.01 dB corner thus becomes approximately −6.02 dB after two passes. Padding limits startup transients but does not recover samples at a discontinuity; edge samples were excluded from comparisons. The

101-tap linear-phase FIR has 50 samples of one-pass group delay, removed by centered symmetric application; its transition-band ripple and attenuation are visible in the measured response plots. A 50/60 Hz notch (candidate Q=30) was not applied because the prespecified line screen did not justify it [1], [35]–[44].

4.3 Feature behaviour across the set

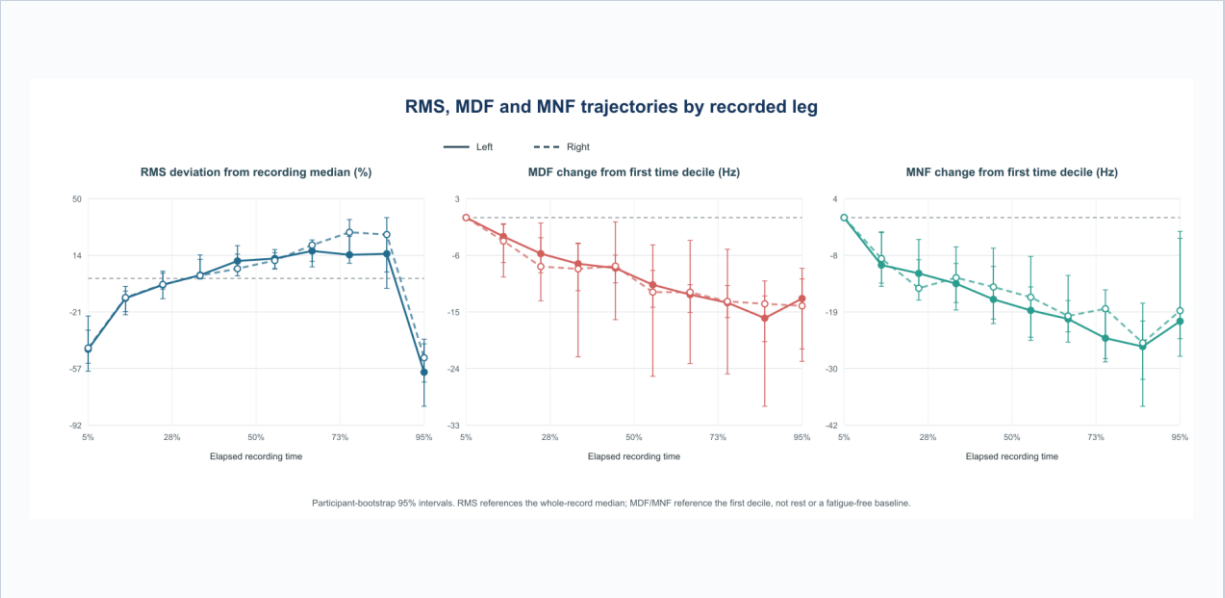


Figure 1. RMS, MDF and MNF trajectories by recorded leg in HP20. RMS is relative to each recording's whole-record median; MDF/MNF differences reference the first time decile. These are not rest baselines or fatigue events.

4.4 RMS, MDF and MNF trajectories by recorded leg

For each complete 1 s window with 50% overlap, the centered waveform is $x_c[n]=x[n]-\bar{x}$. Its DFT is $X[k]=\Sigma_{n=0}^{N-1}x_c[n]w_{Hann}[n]e^{-j2\pi kn/N}$. The one-sided periodogram is $P[k]=c_k|X[k]|^2/(f_s\Sigma_n w_{Hann}[n]^2)$, where c_k doubles interior positive-frequency bins. $RMS=\sqrt{(\Sigma_n x_c[n]^2/N)}$; $MNF=\Sigma_{f=20}^{400}fP(f)/\Sigma P(f)$; MDF satisfies $\Sigma_{20\le f\le MDF}P(f)=0.5\Sigma_{20\le f\le 400}P(f)$, with interpolation between adjacent bins. Frequencies are in Hz, local amplitude in ADC counts and PSD in ADC²/Hz. An ordinary least-squares slope $\beta_1=\Sigma(t_j-\bar{t})(y_j-\bar{y})/\Sigma(t_j-\bar{t})^2$ summarizes descriptor change per minute. A negative β_1 is a direction of change, not an onset detector. No universal percentage or change point was imposed.

4.5 Feature selection or reduction

Post hoc lateral summaries use the global nested selector's already frozen outer predictions. At HP20, the median participant slope for the left leg was RMS +17.02 ADC counts/min (95% CI 8.38 to 41.11; 13/14 positive), MDF −41.66 Hz/min (−77.15 to −22.50; 14/14 negative) and

MNF −43.69 Hz/min (−83.64 to −30.88; 14/14 negative). For the right leg, the slopes were RMS +17.83 ADC counts/min (5.11 to 47.50; 12/15 positive), MDF −38.36 Hz/min (−93.54 to +4.29; 11/15 negative) and MNF −38.99 Hz/min (−85.79 to −9.74; 13/15 negative). The traces are non-monotonic; RMS falls sharply in the final time bin and spectral descriptors rebound. These are time-associated signal trends, not fatigue estimates.

5. Machine learning methodology

5.1 Models evaluated

TABLE VI. ELEVEN MODEL FAMILIES AND INNER-SEARCH SETTINGS

Model family	Role	Hyper-parameters searched	Selection method
Tied-prior baseline	Reference floor	Fixed equal priors 1/3; first-class tie break	Fixed; no fitting
Nearest centroid	Simple distance baseline	Fixed	Grouped inner macro-F1
K-nearest neighbours	Local nonlinear baseline	k ∈ {3, 5, 9}	Grouped inner macro-F1
Ridge classifier	Regularized linear baseline	alpha ∈ {0.1, 1, 10}	Grouped inner macro-F1
Multinomial logistic	Linear probabilistic family	C ∈ {0.01, 0.1, 1}	Grouped inner macro-F1
LDA with shrinkage	Linear discriminant family	shrinkage ∈ {0.1, 0.5, 0.9}	Grouped inner macro-F1
Linear SVM	Large-margin linear classifier	C ∈ {0.1, 1, 10}	Grouped inner macro-F1
Random forest	Small tree ensemble	(8 trees, depth 2) or (12, depth 3); min leaf 2	Grouped inner macro-F1
Gradient-boosted stumps	Shallow boosted ensemble	learning rate 0.1; 6 or 12 estimators	Grouped inner macro-F1
Logistic RMS+MDF	Two-feature linear baseline	RMS and MDF; C ∈ {0.01, 0.1, 1}	Grouped inner macro-F1
LDA RMS+MDF	Two-feature discriminant baseline	RMS and MDF; shrinkage ∈ {0.1, 0.5, 0.9}	Grouped inner macro-F1

5.2 Validation scheme

The candidate predictors comprised 15 time/distribution descriptors, eight spectral descriptors and eight between-window IQR descriptors, evaluated as seven fixed feature-family combinations. The set includes mean, standard deviation, RMS, absolute and median values, quantiles, peak-to-peak, line length, median crossings, skewness, excess kurtosis, MDF, MNF, peak frequency, normalized spectral entropy, relative band powers and IQRs of RMS/spectral quantities. Standardization, model selection and feature-family selection occurred only inside grouped training folds. No global feature selection was fitted across all participants.

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Outer loop: leave one complete participant group out; repeat for all 15 groups.

Training side: the other 14 groups only. All series, left/right recordings and segments from the held-out group stay outside fitting.

Inner loop: four participant-grouped folds select among 325 fixed model × feature-family × filter configurations using training-side macro-F1.

Refit the selected configuration on all 14 outer-training groups; evaluate once on the held-out participant and retain its out-of-fold predictions.

After 15 folds, aggregate the held-out predictions. The outer test fold never selects a model, feature family, filter, threshold or hyper-parameter.

Figure 2. Editable outline of the nested participant-wise selection procedure.

5.3 Class imbalance

Eleven model families were evaluated: tied equal-prior baseline, nearest centroid, K-nearest neighbours, ridge, multinomial logistic regression, shrinkage LDA, linear SVM, random forest, shallow gradient-boosted stumps, logistic RMS+MDF and LDA RMS+MDF. No neural network was trained. Models received window-level features rather than raw waveform end-to-end input. Logistic regression produces nominal class probabilities, but calibration was not assessed; LDA, SVM and other models produce model-specific scores that are not comparable as probabilities.

5.4 Reproducibility settings

TABLE VII. REPRODUCIBILITY SETTINGS AND VERIFIED SCOPE

Item	Value
Outer / inner validation	15 outer LOSO groups; 4 grouped inner folds per outer training set
Random seeds	20261018 nested benchmark; 20261019 lateral summary; see trend receipt for trajectory seed
Bootstrap	5,000 participant-level resamples for primary nested CIs; 2,000 for lateral comparisons; slope/trend intervals as recorded in trend receipt
Software	Python 3.12.14; NumPy 2.3.5; full dependency lock not recorded
Runtime / hardware	31423.1 s recorded; hardware specification not logged
Scope / data integrity	Local recordings from 15 groups only; participants.csv and external data excluded; originals, labels, partitions and 44 references unchanged
Configuration / code receipt	325 configurations; receipt completed; run 20261017_224915; participant grouping verified

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Validation used 15 outer leave-one-participant-out folds. Each outer test group contained that participant's signals, both sides, all R1–R5 series and associated segments. Four group-aware inner folds selected among 325 fixed model × feature-family × filter configurations, then the selected configuration was refit on the other 14 groups and evaluated once on the held-out group. The global nested procedure is a selector that can vary by fold; it is not one frozen deployable model. Side and series were not used to select the model. Participant-level bootstrap intervals used 5,000 resamples for the global nested primary summary and 2,000 for the post hoc lateral comparison.

6. Results

6.1 Main comparison table

TABLE VIII. PARTICIPANT-LEVEL PERFORMANCE OF EACH NESTED SELECTOR

Selector	Accuracy mean±SD	Precision macro	Recall macro	Macro-F1	ROC-AUC OvR / AP OvR
Tied-prior baseline	0.333±0.000	0.111±0.000	0.333±0.000	0.167±0.000	0.500±0.000 / 0.333±0.000
Nearest centroid	0.555±0.121	0.540±0.182	0.555±0.121	0.482±0.152	0.643±0.069 / 0.568±0.061
KNN	0.566±0.115	0.633±0.141	0.566±0.115	0.551±0.116	0.782±0.094 / 0.654±0.119
Ridge classifier	0.734±0.105	0.794±0.090	0.734±0.105	0.716±0.119	0.946±0.045 / 0.913±0.069
Logistic regression	0.764±0.099	0.824±0.077	0.764±0.099	0.751±0.108	0.945±0.046 / 0.914±0.078
LDA	0.746±0.096	0.807±0.083	0.746±0.096	0.730±0.108	0.934±0.045 / 0.892±0.070
Linear SVM	0.741±0.102	0.804±0.077	0.741±0.102	0.726±0.113	0.945±0.046 / 0.911±0.077
Random forest	0.646±0.166	0.702±0.167	0.646±0.166	0.611±0.200	0.898±0.067 / 0.847±0.096
Gradient-boosted stumps	0.540±0.144	0.511±0.218	0.540±0.144	0.464±0.179	0.760±0.112 / 0.601±0.122
Logistic (RMS+MDF)	0.532±0.146	0.522±0.186	0.532±0.146	0.454±0.172	0.814±0.091 / 0.725±0.123
LDA (RMS+MDF)	0.548±0.145	0.563±0.172	0.548±0.145	0.476±0.176	0.803±0.071 / 0.704±0.087
Global nested selector	0.755±0.088	0.810±0.074	0.755±0.088	0.740±0.099	0.941±0.038 / 0.906±0.064

Note: ROC-AUC and average precision use each fold's continuous class scores (probabilities or model-specific discriminant/margin scores). They are not evidence of probability calibration. Calibration scores are not pooled for the global selector because its selected folds mix logistic probabilities and LDA discriminant scores.

The current run receipt records completion, 15 outer folds, four grouped inner folds, 325 fixed configurations, 11 model families, seven feature-family combinations, five filter branches, 414 selected segments, seed 20261018 and 5,000 participant bootstrap replicates. The receipt reports no use of participant.csv, external data, external compute, changes to signals, frozen labels or partition manifest. Python 3.12.14 and NumPy 2.3.5 were recorded; hardware specifications and a complete dependency lock were not. The local run and output hashes are retained in the phase record.

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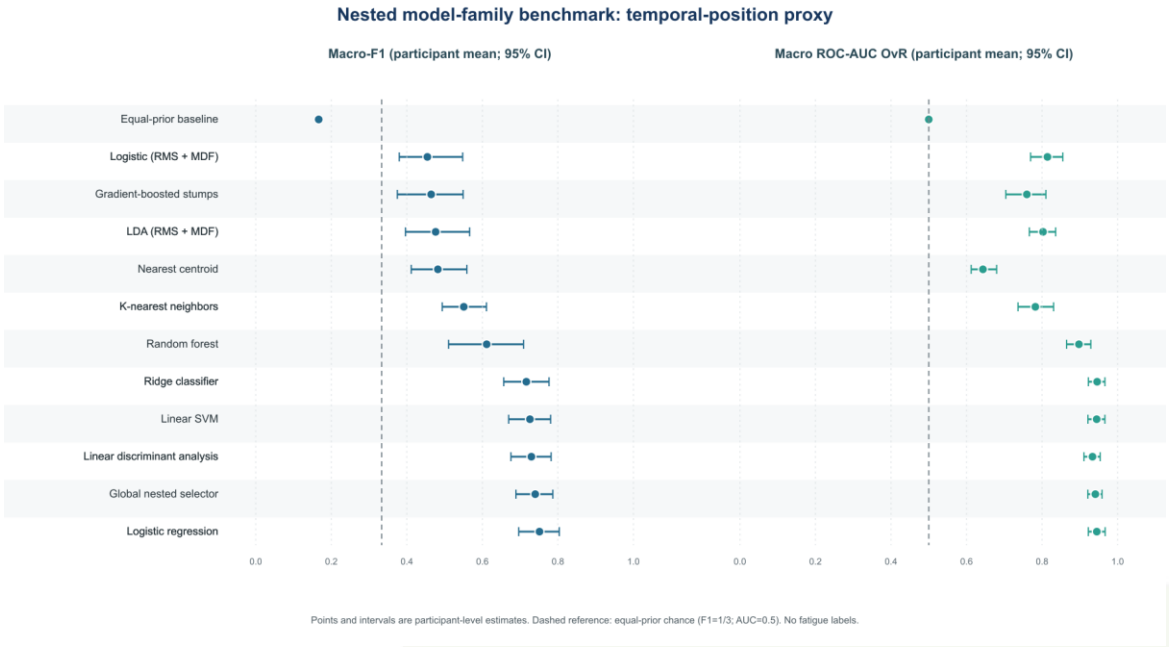


Figure 3. Participant-level macro-F1 and macro ROC-AUC for all fixed model-family selectors and the global nested procedure, with 95% participant-bootstrap intervals. Dashed references show equal-prior chance; labels are temporal thirds.

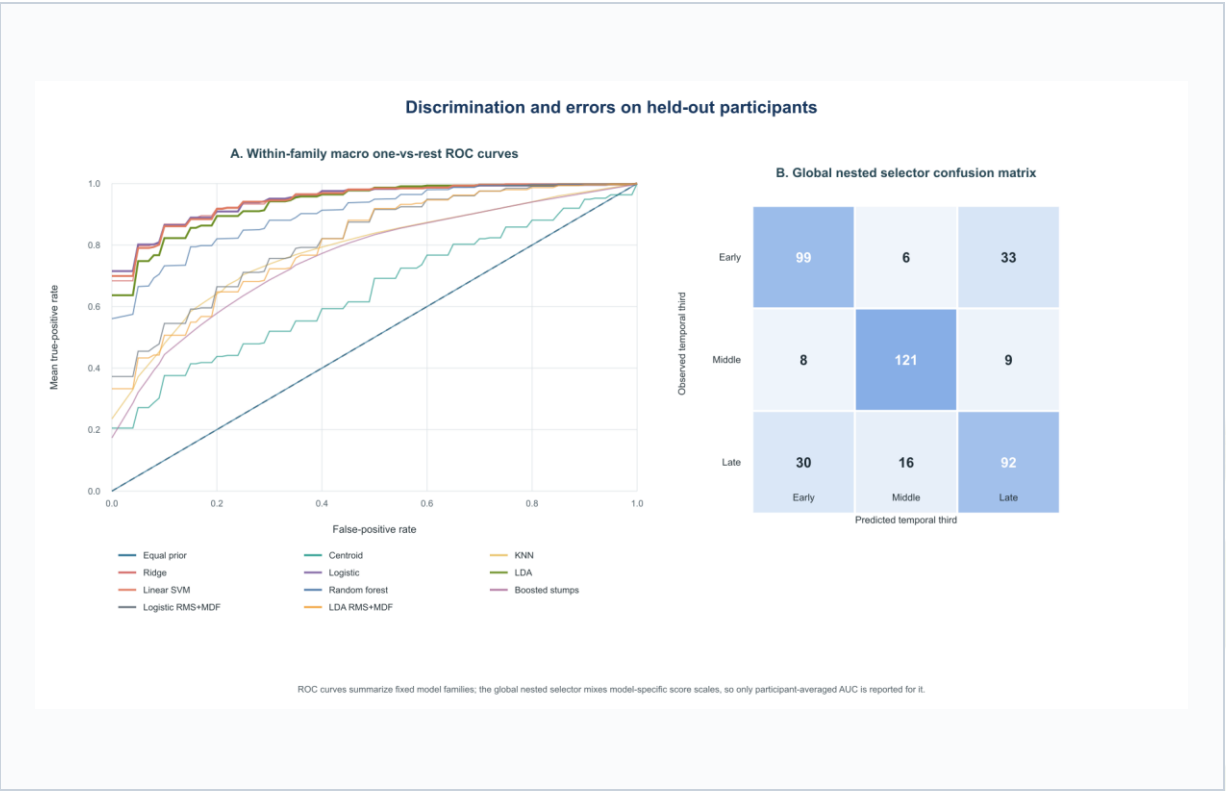
6.2 Pooled per-class performance of the global nested selector

TABLE IX. POOLED OUT-OF-FOLD PER-CLASS METRICS

Proxy class	Precision	Recall	F1	Support
Early	0.723	0.717	0.720	138
Middle	0.846	0.877	0.861	138
Late	0.687	0.667	0.676	138
Macro (pooled counts)	0.752	0.754	0.752	414

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6.3 Confusion matrix



Across participants, the global nested selector achieved accuracy 0.755±0.088, macro precision 0.810±0.074, macro recall/balanced accuracy 0.755±0.088, macro-F1 0.740±0.099 (95% participant-bootstrap CI 0.689–0.787), ROC-AUC OvR 0.941±0.038 (CI 0.921–0.959), average precision OvR 0.906±0.064 (CI 0.873–0.936), MCC 0.665 and Cohen's κ 0.632. Among the fixed model-family selectors, logistic regression had the largest point macro-F1, 0.751±0.108 (CI 0.696–0.804), and ROC-AUC 0.945±0.046 (CI 0.923–0.967). Its interval overlaps the nested selector and other strong linear families; no superiority test was performed. The baseline predicts the first class under equal priors and achieves macro-F1 0.167 and AUC 0.500.

Figure 4. Family-level macro one-vs-rest ROC curves and the global nested selector confusion matrix for elapsed-time thirds, not fatigue.

Pooled nested out-of-fold counts were 414 segments, with 138 support per class. The confusion matrix (rows observed, columns predicted) was [[99,6,33],[8,121,9],[30,16,92]]. Pooled class F1 was 0.720 early, 0.861 middle and 0.676 late; the corresponding pooled precision was 0.723, 0.846 and 0.687 and recall was 0.717, 0.877 and 0.667. These pooled values are computed from counts and differ from participant-averaged metrics because F1 is nonlinear. Errors between early and late positions (33 early→late; 30 late→early) were more frequent than adjacent-bin errors.

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6.4 Leg-by-series proxy-performance summary



Figure 5. Nested-selector performance by recorded leg and series; coverage differs across sides and R1–R5.

The heatmap summarizes the post hoc leg-by-series temporal-proxy scores. Coverage is 14 participants per left-leg series and 15, 14, 13, 13 and 13 for right R1–R5. The paired R1–R5 contrasts use 14, 13, 12, 12 and 12 participants. Intervals are wide and no side or series was used to select a model; these estimates do not demonstrate fatigue asymmetry.

6.5 Bilateral and series-level comparison

The post hoc side analysis included 14 left-leg participants (70 files, 210 segments) and 15 right-leg participants (68 files, 204 segments); 14 participants were paired across sides. Nested-selector macro-F1 was 0.776 (95% CI 0.675–0.874) left and 0.686 (0.611–0.762) right. Paired left-minus-right macro-F1 was +0.092 (CI −0.052 to +0.238); paired differences for balanced accuracy, AUC, AP and MCC also included zero. Across series, left-side F1 was 0.754, 0.730, 0.651, 0.750 and 0.841 for R1–R5. Right-side F1 was 0.567, 0.655, 0.568, 0.585 and 0.739; coverage ranged from 15 participants in right R1 to 13 in R3–R5. Paired per-series comparisons used 14, 13, 12, 12 and 12 participants, respectively, and are exploratory. The heatmap and tables describe temporal-position performance only.

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7. Discussion

7.1 Interpretation

The principal result is that a grouped selection procedure can classify elapsed position above an equal-prior baseline in this local corpus. It does not establish that the model detects peripheral or central fatigue. The class labels are defined by time, and the signal may reflect contraction density, pauses, changing effort, movement, file duration or acquisition drift. The high participant-averaged ROC-AUC and lower macro-F1 reflect distinct aspects of ranking and decision errors and do not convert the target into fatigue. A current nested selector that changes between logistic regression and LDA also cannot be called one frozen model for deployment.

7.2 Comparison with the literature

The direction of the local MDF/MNF changes is compatible with literature on some sustained contractions under approximately constant force [31], [32], but this dataset has no verified force trace or aligned contraction phase. The low-frequency trend estimates, nonlinear trajectories, final RMS decline and spectral rebounds are consistent with multiple possible causes. Dynamic-fatigue studies comparing Fourier and wavelet indices have reported variable validity and reliability [34]; this supports examining time-frequency behavior but does not make a wavelet or a negative slope a physiological reference. Similarly, AI studies use Borg ratings, power loss, anaerobic thresholds, sensors or repeated-session calibration [5]–[7], [11], [13], [16], [17], [19]–[27], [33]. Their definitions and participant splits differ from the local time-third task.

7.3 Failure analysis

The errors most often confuse the first and final time thirds, whereas the transition third is more frequently correct in the pooled matrix. Without per-repetition labels, force and task phase, no causal explanation can be assigned to individual errors. The ROC plot and confusion matrix quantify only temporal-third discrimination. They are not fatigue false-positive/negative counts, sensitivity or specificity.

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7.4 Limitations

The sample comprises 15 groups with incomplete side-series coverage. RAW/RECT/ENV mode, electrode placement, muscle identity, analog chain, ADC saturation rails, calibrated voltage, force, load, repetitions, cadence, pauses and failure time are unverified. Signal discontinuities were split rather than interpolated; QC review flags are not a substitute for a clean reference. There is no local physiological fatigue ground truth, no external validation in the current local-only scope and no interventional randomization. ROC and F1 uncertainty is substantial for a 15-participant study.

7.5 What you would do differently

A follow-up study should document the muscle and electrode montage, acquisition chain, analog and digital filters, calibrated sampling and voltage, load, cadence, rests, and repetition-level event markers. It should synchronously record force or torque and predefine an independent fatigue criterion, then evaluate subjects with grouped splits and a frozen model/threshold. Baseline and late windows should represent comparable contraction phases. External validation may be considered only as a separately approved, prespecified future phase; it was not performed here.

8. Conclusions

1) The QC audit covered 140 recordings and 4,256,257 samples from 15 groups. It found no structural exclusions; timestamp gaps were not interpolated and R0/R0_1 remained QC-only.
2) The 325-configuration nested LOSO procedure achieved temporal-third macro-F1 0.740±0.099 and ROC-AUC OvR 0.941±0.038. Logistic regression had the best fixed-family point macro-F1 (0.751±0.108), but no superiority was established and the nested selector is not one frozen model.
3) The bilateral analysis covered 14 left and 15 right participants. Paired side differences included zero, while per-series coverage was unequal; no physiological leg asymmetry is demonstrated.
4) HP20 time-associated slopes had decreasing MDF/MNF and increasing median RMS on both sides, but the trajectories were non-monotonic and not anchored to force or verified fatigue events.
5) The findings support only a temporal-position benchmark. No fatigue detector, fatigue threshold, onset, peak, sensitivity, specificity, external validation or fatigue FP/FN rate is claimed.

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9. Contribution of each member

TABLE X. CONTRIBUTIONS REQUIRE AUTHOR CONFIRMATION

Member / role	Contribution evidence	Approx. effort
Author team (identities not in audited project record)	Analyses and drafting require author attribution/confirmation	Not recorded; author team to declare
Data acquisition / consent	No attributable member-level record in reviewed materials	Not recorded
Code / review / supervision	No attributable member-level record in reviewed materials	Not recorded
Funding / institutional approvals	Not identified in audited files	Not recorded

Ethics and consent: the audited materials did not contain an approval or consent record. The responsible authors must confirm the applicable institutional documentation before submission.

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Appendix A - Additional figures

Appendix A contains the aggregate bilateral RMS/MDF/MNF trajectories, nested model comparison, family ROC curves, nested confusion matrix, leg-by-series heatmap, filter responses and aggregate filter effects. These figures supplement the main analysis and do not establish physiological fatigue.

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Figure A1. Side-stratified RMS, MDF and MNF trajectories across normalized recording time. Bands are participant-bootstrap intervals; the first time decile is not a fatigue-free baseline.

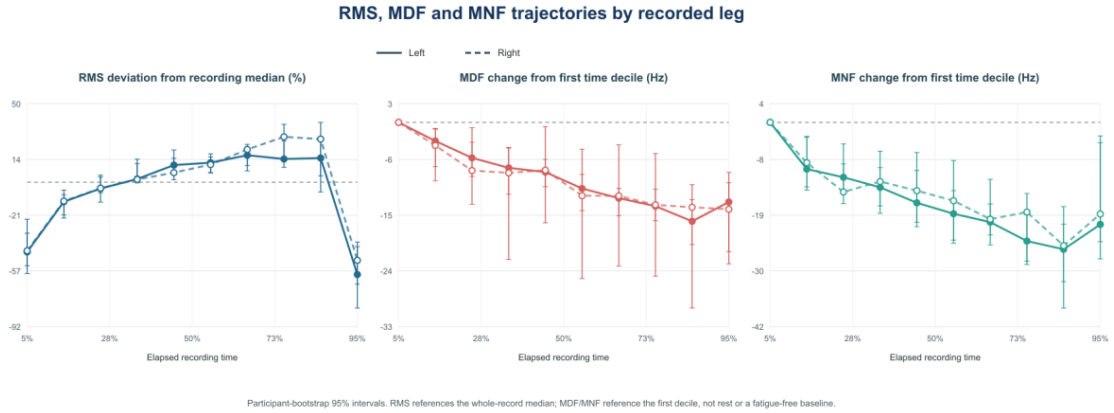
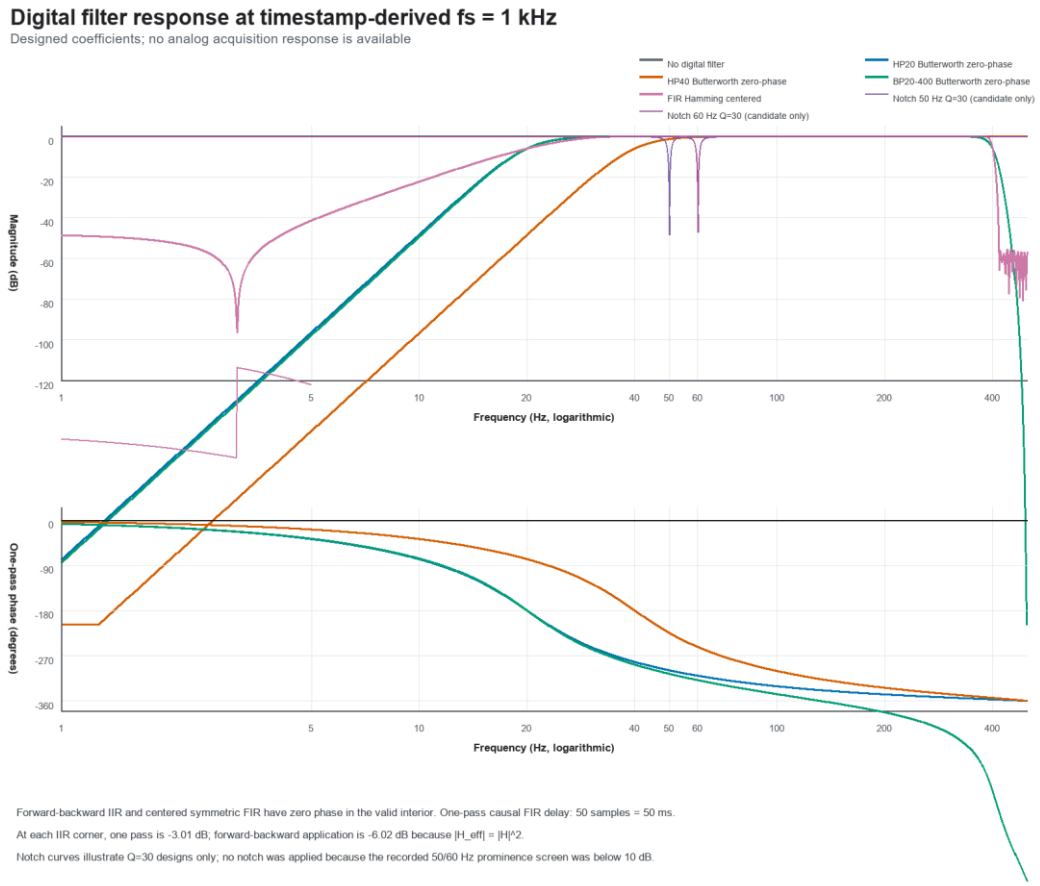
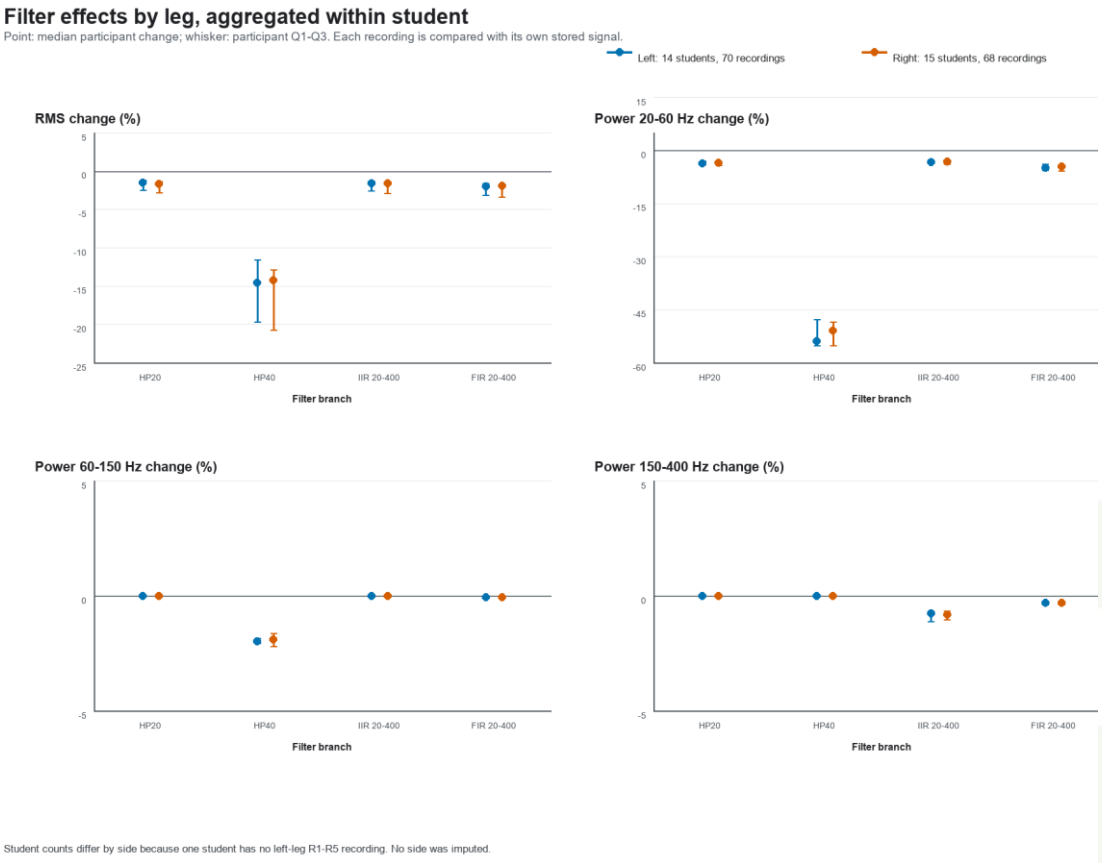


Figure A2. Candidate-filter magnitude and phase responses, including one-pass and forward–backward IIR behavior. The plot is the designed digital response, not a measured clean-signal reference.



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Figure A3. Quantitative change in RMS and band power after filtering, summarized by recorded leg. These changes show what the filters alter; they do not estimate SNR or identify a denoising winner.



Reference filter coefficient arrays at 1 kHz are supplied in COEFICIENTES_REFERENCIA_FILTROS_F31_fs1000Hz.csv. The benchmark recomputed the coefficients from each contiguous run's timestamp-derived sampling rate; the CSV is a reproducible design reference, not a substitute for those run-specific arrays.

Appendix B - Full feature list

Appendix B enumerates all 31 model descriptors and the seven feature-family combinations. Units and operational definitions are summarized in Table V. These candidate predictors are not validated fatigue markers.

Time-domain (15): adc_mean, adc_sd_population, adc_rms, adc_mean_absolute, adc_median, adc_median_absolute, adc_p05, adc_p25, adc_p75, adc_p95, adc_peak_to_peak, adc_line_length_mean, adc_median_crossing_rate, adc_skewness_population, adc_excess_kurtosis_population

Spectral (8): median_frequency_hz, mean_frequency_hz, peak_frequency_hz, spectral_entropy_norm, relative_power_20_60, relative_power_60_150, relative_power_150_250, relative_power_250_400

Between-window IQR (8): rms_iqr, median_frequency_iqr_hz, mean_frequency_iqr_hz, spectral_entropy_iqr_norm, relative_power_iqr_20_60, relative_power_iqr_60_150, relative_power_iqr_150_250, relative_power_iqr_250_400

The seven combinations are time, frequency, between-window variability and the three pairwise combinations plus all three. The two-feature RMS+MDF baselines are separate specialist representations. All feature-family selection remained within grouped inner folds.

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Appendix C - Repository map

The repository map below identifies the analysis code, aggregate outputs, report figures and project-tracking files. Private row-level predictions and original EMG recordings are not embedded in the report.

datasets/codigo_local/ — signal QC, filters, temporal labels, feature extraction, nested model search, ablation and verification scripts.

datasets/derivados/ — aggregate-only QC, model/family/filter metrics, confusion matrices, trend estimates and feature-ablation summaries.

articulo/figuras/reporte_F31_20261018/ — aggregate model comparison, ROC/confusion and bilateral trajectory plots; no individual participant plot is included.

articulo/gestion/ — phase log, decisions, claim/compliance matrices, audit notes, receipts and reproducibility workbook.

articulo/entregables/ — F31 IEEE report, source manuscript and reference filter coefficient CSV. Private row-level out-of-fold outputs remain in local application data and are not embedded in the report.

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